

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
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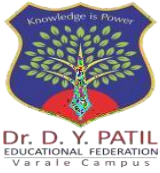
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AI LOAN ELIGIBILITY CHECKER: AI-Powered BFSI Decision-Support Platform

Project Description :-

AI Loan Eligibility Checker is an AI-assisted financial decision-support platform designed for the Banking, Financial Services and Insurance (BFSI) domain. The platform helps users understand their indicative loan readiness by analyzing important financial parameters such as monthly income, age, employment type, work experience, credit score, existing EMIs, loan type, requested loan amount and repayment tenure.

The system uses transparent, rule-based calculations to provide an educational eligibility result instead of making an actual lending decision. Users receive an eligibility category such as Eligible, Partially Eligible or Not Eligible, along with an educational eligibility score, estimated maximum loan amount, estimated



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interest-rate range, FOIR and reasons affecting the result.

The platform also provides a Credit Score Analyzer that estimates a user's credit health on a 300–900 scale using factors such as payment history, credit utilization, credit age, active loans, hard inquiries and credit mix. Users can understand which financial habits may affect their estimated credit profile.

An integrated EMI Planner allows users to calculate monthly EMI, total interest and total repayment for a selected loan amount, interest rate and tenure. The system also provides an amortization schedule and a visual representation of principal and interest.

The project includes an AI-powered Financial Tips section that can provide personalized educational guidance related to saving, debt reduction, credit scores, investing basics and insurance. Claude can be used for AI-generated explanations, while rule-based fallback guidance is available when the AI service is unavailable.

The application also provides Saved Records functionality using Google Sheets integration with browser local-storage backup. The interface follows a responsive glassmorphism-inspired design and is designed to work across desktop and mobile devices.

The project is implemented using HTML5, CSS3, JavaScript, Chart.js, Node.js/serverless API adapters, Claude API integration, Google Apps Script and Google Sheets. The application is available as a live web application.

Technology & AI :-

Frontend: HTML5 / CSS3 / Vanilla JavaScript / Responsive UI / Chart.js

Backend & API: JavaScript / Serverless API Functions / Claude API Proxy

Data Storage: Browser Local Storage / Google Sheets API / Google Apps Script

AI: Anthropic Claude API / Rule-Based Financial Tips Fallback

Financial Calculations: Loan eligibility rules / FOIR calculation / EMI calculation / Amortization schedule / Credit-score estimation

Security: HTTPS deployment / Environment Variables for API Keys / Input Validation / Safe Text

Rendering / API Request Controls

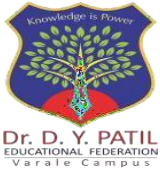
Deployment: Netlify / Vercel compatible deployment configuration

Version Control: Git / GitHub

Architecture: Static Single-Page Application with serverless AI integration

Key Features :-

- **AI-assisted loan eligibility checking**
- **Eligibility result categories: Eligible, Partially Eligible and Not Eligible**
- **Educational eligibility score from 0–100**
- **Estimated maximum loan amount**
- **Estimated interest-rate range**
- **FOIR calculation and affordability analysis**
- **Analysis based on income, employment, obligations and credit signals**
- **Credit Score Analyzer using a 300–900 estimated score range**
- **Credit analysis using payment history, credit utilization, credit age, active loans, hard inquiries and credit mix**
- **Risk-level and factor-wise credit analysis**
- **Interactive EMI calculator**
- **Loan amount, interest-rate and tenure controls**
- **Monthly EMI calculation**
- **Total interest and total repayment calculation**
- **Principal versus interest visualization using a doughnut chart**
- **Complete amortization schedule**



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AI-powered financial tips

- **Financial guidance for saving, debt reduction, credit score, investing basics and insurance**
- **Optional Claude AI explanations and follow-up questions**
- **Rule-based fallback when AI service is unavailable**
- **Saved eligibility records**
- **Google Sheets integration for storing records**
- **Local browser storage backup when Google Sheets is unavailable**
- **Responsive mobile and desktop interface**
- **Inline form validation and user-friendly error handling**
- **Privacy-focused design without requesting passwords or government ID numbers**
- **Educational disclaimer to clarify that the system does not guarantee loan approval**

Scenarios :-

Scenario 1: A user wants to know whether they may be ready for a loan. The user enters their name, income, employment type, age, experience, credit score, existing EMIs, loan type, requested amount and tenure. The system analyzes these inputs and displays an educational eligibility result, eligibility score, estimated maximum amount, estimated rate range, FOIR and factors affecting the result.

Scenario 2: A user wants to understand their credit health. The user enters information about payment history, credit utilization, credit age, active loans, hard inquiries and credit mix. The Credit Score Analyzer estimates a score within the 300–900 range and displays factor-wise information and risk indicators.

Scenario 3: A borrower wants to calculate the affordability of a loan. The user enters the loan amount, annual interest rate and repayment tenure. The EMI Planner calculates the monthly EMI, total interest and total repayment and provides an amortization schedule. The principal and interest portions are also displayed visually.

Scenario 4: A user wants financial guidance after checking their loan position. The user selects a topic such

as saving, debt reduction, credit score, investing basics or insurance and provides optional income and expense information. The AI financial assistant can provide educational and personalized suggestions.

Scenario 5: A user wants to maintain a history of previous eligibility checks. After completing an eligibility evaluation, the user can save the result. Records can be maintained through Google Sheets when configured, with local browser storage available as a backup.

Milestones :-

Milestone 1 – Project Planning & Requirement Analysis

Define the objective of the AI Loan Eligibility Checker, identify the target BFSI users, determine the financial inputs required for eligibility analysis and plan the major modules of the application.

Milestone 2 – Frontend & User Interface Development

Develop the responsive single-page interface using HTML5, CSS3 and JavaScript. Create sections for loan eligibility, credit health, EMI planning, financial tips and saved records.

Milestone 3 – Loan Eligibility & Financial Calculation Module

Implement the eligibility form, financial validation, eligibility scoring, affordability analysis, FOIR calculation, estimated loan amount and result explanations.

Milestone 4 – Credit Health & EMI Modules

Develop the credit score estimation system using multiple credit factors and implement the EMI calculator, interest calculation, repayment calculation and amortization schedule.

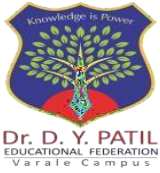
Milestone 5 – AI Assistant & Data Integration

Integrate the Claude API through a secure serverless proxy and implement rule-based fallback guidance. Add Google Sheets integration and local-storage backup for saved eligibility records.

Milestone 6 – Testing, Optimization & Deployment

Verify JavaScript syntax, JSON configuration, application routes, calculations, AI fallback behavior, saved records and responsive layouts. Configure Netlify/Vercel deployment and verify the live application.

Conclusion :-



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AI Loan Eligibility Checker provides an interactive and transparent way for users to understand their loan readiness and financial position. By combining rule-based eligibility analysis, credit health estimation, EMI planning, financial calculations and optional Generative AI assistance, the platform converts complex financial information into an easier-to-understand educational experience.

The project demonstrates the practical use of modern web technologies such as HTML5, CSS3, JavaScript, Chart.js, serverless APIs, Anthropic Claude, Google Sheets and GitHub to develop a BFSI-focused financial technology application.

The system is designed as a decision-support and educational platform rather than a replacement for professional financial advice or actual lender underwriting.

Expected Outcome :-

AI Loan Eligibility Checker aims to help users understand their approximate loan readiness, financial affordability and credit-health factors before approaching a lender.

The platform provides an integrated environment for eligibility analysis, credit-score estimation, EMI calculation, amortization planning, financial tips and saved records.

The project demonstrates practical knowledge of Artificial Intelligence, Generative AI, BFSI applications, frontend development, JavaScript, financial calculations, data visualization, REST/serverless APIs, Google Sheets integration, responsive design, GitHub and cloud deployment.

The project can serve as a foundation for future enhancements such as user authentication, private multi-device records, PDF report generation, multilingual support and ML-based prediction models trained with lender-approved data.